



ECON 100B: Macroeconomics

Class 25: Exchange Rates & International Finance (Jones Chapter 20)

Ryan D. Edwards
April 28, 2026

Agenda

- Announcements
- Current events
- Chapter 20: Exchange Rates

Announcements

- **Problem Set 4** is available and is due on **Friday, May 1**
- All students get to drop their lowest PS score. If you have scored 100% on PS1, PS2, and PS3, you can skip PS4. But I don't recommend it
- **Review session** here in class on Thursday, April 30. Please bring questions
- There are now two previous final exams available on Drive: Spring 2011 and Spring 2014. (Note that Spring 2011 was 120 minutes)
- The **Final exam** is confirmed for **Monday, May 11**
 - Just like at midterm, the main class meets here in **Pimentel 1**
 - We will run 90 minutes from about 3:00-4:30PM
 - DSP students: Please either make a proctoring appointment, or for 150% time only, we will have **Anthropology and Art Practice 160**

(I think it's kind of near Caffè Strada on Bancroft)

Current events

- [Per WSJ](#): On Friday, the [U.S. Attorney in DC announced on X](#)
 - That she directed her office to close its investigation into the Fed's Eccles Building renovation
 - Pirro referenced the inquiry by the Fed's Inspector General, [begun by Chair Powell in July 2025](#), and wrote, "I will not hesitate to restart a criminal investigation should the facts warrant doing so."
- [Per WSJ](#): On Sunday, Sen Thom Tillis (R-NC) [wrote on X](#)
 - Justice staff clarified privately what they meant
 - Tillis was dropping his hold on Kevin Warsh's nomination to succeed Powell
 - "Only a criminal referral from the inspector general would cause a reopening of the investigation."
- Fingers crossed

Learning objectives for Chapter 20 [1/2]

- A country's nominal exchange rate E is the **price** of its domestic currency: foreign currency per unit of domestic currency, like € 0.85 euros per U.S. dollar \$1.00
- In the long run, any nominal exchange rate E should reflect the **Law of One Price** that connects E to price levels

$$E = \frac{P^w}{P} \quad \text{which means} \quad \begin{aligned} g[E] &= g[P^w] - g[P] \\ g[E] &= \pi^w - \pi \end{aligned}$$

If Euro-area inflation $\pi^w = 0.02$ and U.S. inflation $\pi = 0.02$, then we would expect $g[E] = 0$

- Not all currency areas have a 2% inflation target, and one would expect currencies with faster inflation to **depreciate** against the dollar and against the euro

Learning objectives for Chapter 20 [2/2]

- In the short run, prices and price levels are usually sticky, while nominal exchange rates E can fluctuate quickly. (Think about planning a trip: E jumps a lot, and it can induce anxiety)
- What causes E to fluctuate? Demand & supply affect the currency's price
 - Changes in the domestic interest rate R or the foreign rate R^w trigger shifts in pools of savings seeking higher returns, and thus demand for the currency
 - Uncertainty, fear, and “flight to safety,” usually toward US\$; or away? We will see
- Jumps in E can increase or reduce net exports and thus $\tilde{Y}$
 - Rising E means an appreciation; currency buys more; that reduces $NX = EX - IM$
 - Falling E means a depreciation; currency buys less; that increases $NX = EX - IM$
- When ΔE is caused by ΔR , that is a *movement along a flatter IS curve*
When ΔE is caused by something else, that is a *shift in the IS curve*

Lots of exchange rates

<https://www.wsj.com/market-data/currencies/exchangerates>

Unpacking exchange rates

- A country's nominal exchange rate E is the price of its domestic currency: foreign currency exchanged per unit of domestic currency. For the U.S.,

- 160 Japanese yen per U.S. \$	160 $\frac{JP¥}{\$}$
- 6.84 Chinese yuan renminbi per U.S. \$	6.84 $\frac{CN¥}{\$}$
- 0.74 British pound sterling per U.S. \$	0.74 $\frac{£}{\$}$ or $\frac{GBP}{\$}$

- Imagine a good that is traded everywhere in the world, without transportation costs or taxes, like a smartphone game
 - If markets can freely buy and sell the game, then it should retail for “one price,” meaning the same price translated across all currencies, before any fees
 - Suppose its U.S. price was \$1. Then with the exchange rates above, its prices in JP, CN, and GB should be JP¥ 160, CN¥ 6.84, and £ 0.74

Unpacking exchange rates

- A country's nominal exchange rate E is the price of its domestic currency: foreign currency exchanged per unit of domestic currency. For the U.S.,

- 160 Japanese yen per U.S. \$ $160 \frac{JP¥}{\$}$
- 6.84 Chinese yuan renminbi per U.S. \$ $6.84 \frac{CN¥}{\$}$
- 0.74 British pound sterling per U.S. \$ $0.74 \frac{£}{\$} \text{ or } \frac{GBP}{\$}$

- Imagine a good that is traded everywhere in the world, without transportation costs or taxes, like a smartphone game
 - If markets can freely buy and sell the game, then it should retail for “one price,” meaning the same price translated across all currencies, before any fees
 - Suppose its U.S. price was \$1. Then with the exchange rates above, its prices in JP, CN, and GB should be JP¥ 160, CN¥ 6.84, and £ 0.74

The Law of One Price

- The Law of One Price implies that when there is a lot of free trade between economies,

$$E = \frac{P^w}{P}$$

The exchange rate should equal the ratio of prices, *and of price levels*

“World” price P^w = foreign currency price

- Recall our simple smartphone game that costs $P = \$1$

$$E_{JP¥} = \frac{JP¥160}{\$1} = 160$$

- Now suppose the price in the U.S. remained unchanged at \$1, while its price in Japan increased 10%, to ¥176 = 160 + 16
- If trade in the game was really important, you would expect the exchange rate also to increase 10% to 176 ¥/\$. The mechanism would be increased demand for the game purchased in dollars, hence increased demand for dollars, an increased price of dollars $\Delta E > 0$

The Law of One Price

- The Law of One Price implies that when there is a lot of free trade between economies,

$$E = \frac{P^w}{P}$$

The exchange rate should equal the ratio of prices, *and of price levels*

- Recall our simple smartphone game that costs $P = \$1$

$$E_{JP¥} = \frac{JP¥160}{\$1} = 160$$

- Now suppose the price in the U.S. remained unchanged at \$1, while its price in Japan increased 10%, to ¥176 = 160 + 16
- If trade in the game was really important, you would expect the exchange rate also to increase 10% to 176 ¥/\$. The mechanism would be increased demand for the game purchased in dollars, hence increased demand for dollars, an increased price of dollars $\Delta E > 0$

But lots of goods and services are not traded internationally

TABLE 20.1

The Big Mac Index

	Big Mac price in local currency	Exchange rate per dollar (\$)	Big Mac price in dollars
United States	5.15 dollars	1.00 dollars/\$	5.15
Norway	62.00 kroner	9.90 kroner/\$	6.26
Euro area	4.65 euros	0.98 euros/\$	4.74
Mexico	70.00 pesos	20.41 pesos/\$	3.43
China	24.00 yuan	6.75 yuan/\$	3.56
Japan	390.00 yen	137.87 yen/\$	2.83
South Africa	39.90 rand	17.04 rand/\$	2.34
India	191.00 rupees	79.95 rupees/\$	2.39

Source: [economist.com/big-mac-index](https://www.economist.com/big-mac-index) (January 2023).

Source: [economist.com/big-mac-index](https://www.economist.com/big-mac-index) (January 2023).

Then what use is this really?

- The Law of One Price provides a useful way to think about long-run exchange rate dynamics

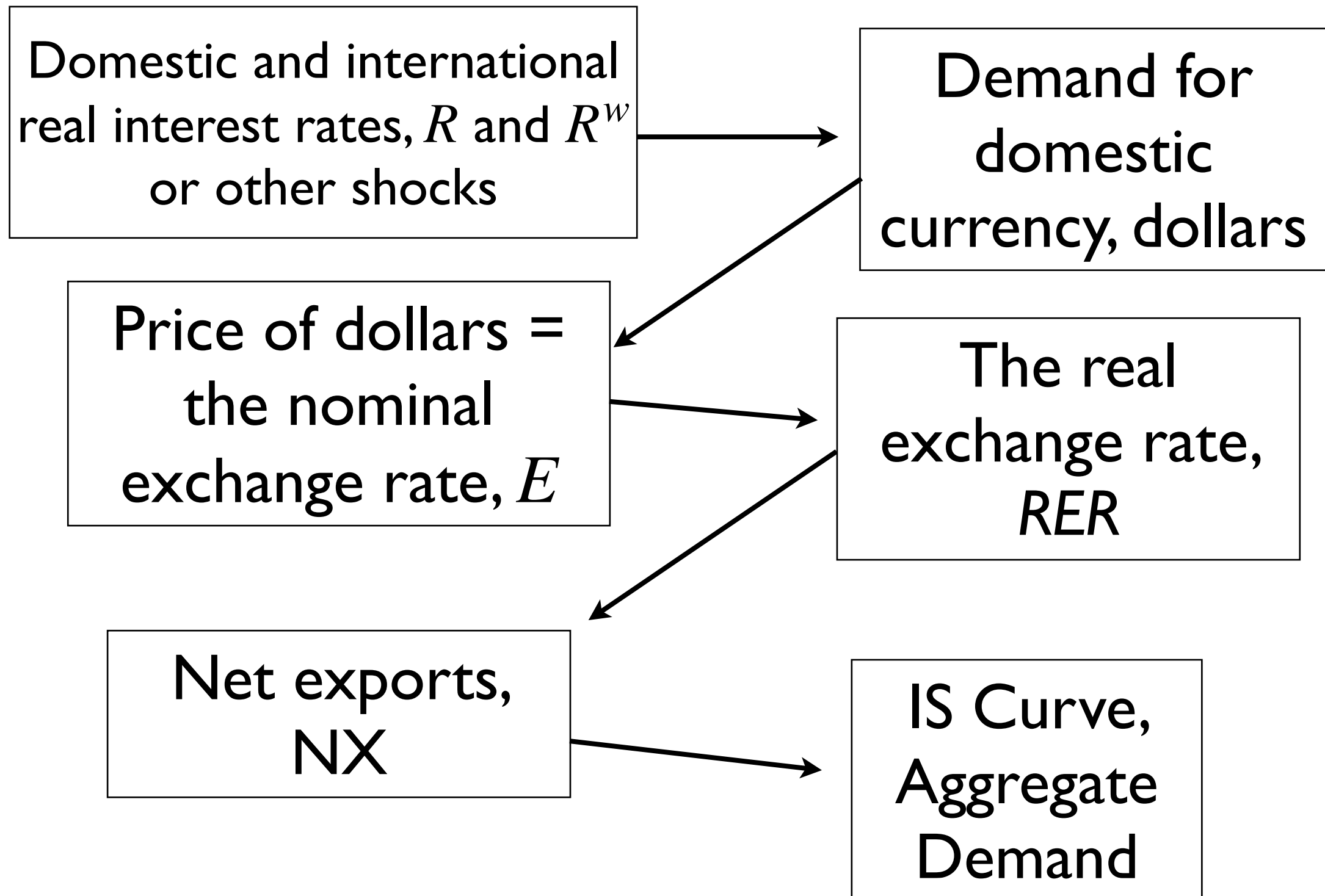
$$E = \frac{P^w}{P} \text{ then } \quad g[E] = g[P^w] - g[P]$$
$$g[E] = \pi^w - \pi$$

P^w is a foreign country's price level and π^w is its rate of inflation

- Central banks in many developed countries have publicly stated their inflation target is $\bar{\pi} = 0.02$
- Some developing countries have different inflation targets. The target of the [Reserve Bank of India](#) is $\bar{\pi}_{IN} = 0.04$
- If central banks maintain their inflation targets, one would expect exchange rates between developed countries to be fairly stable, while the U.S. dollar and other currencies might appreciate against the INR, ₹

Exchange rates and $\tilde{Y}$

Conceptual roadmap

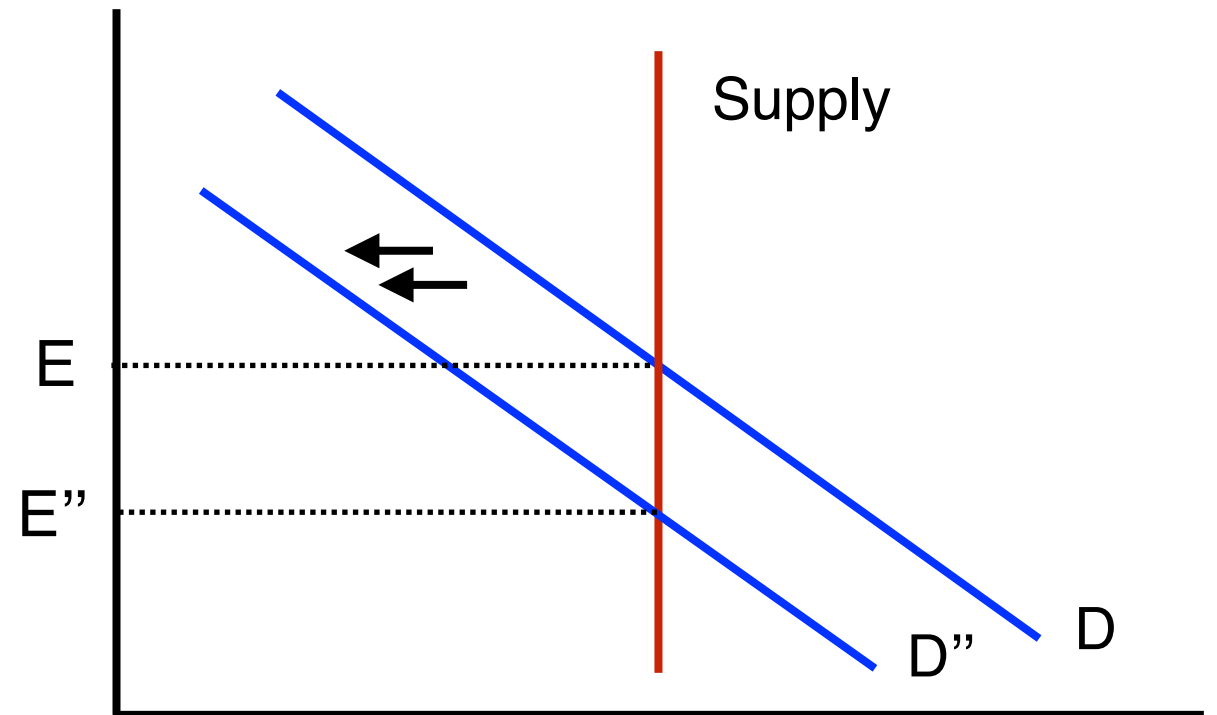
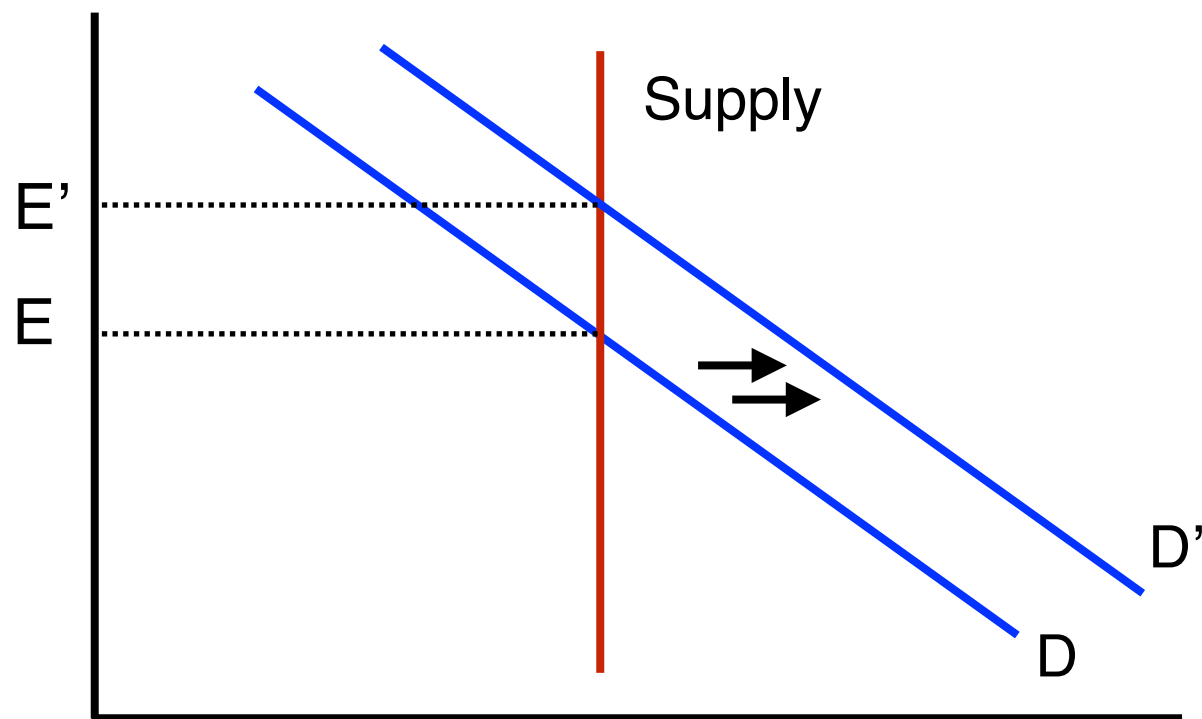


Why do R and R^w affect the demand for dollars?

- Think about an American, German, Japanese, Chinese, or other saver who wants to earn a return on their savings
- In the long run, the marginal product of capital should be the same in different countries with similar credit markets and economic growth, so maybe the saver is ambivalent about where to invest
- But imagine that the U.S. real interest rate R rises above the U.S. marginal product of capital $\bar{r}$, perhaps because the Fed is fighting U.S. inflation
- What should the saver do? Buy U.S. assets, since they pay a higher R
- But in order to buy more U.S. assets, the saver needs to translate savings into dollars, so they demand dollars from the local bank, *raising the demand for dollars*
- So a rise in the U.S. real interest rate R raises the demand for dollars because it attracts more foreign savings

Shifting demand for currencies changes their prices and exchange rates

- Think of the **supply** curves of currencies as being fixed and unchanging
- Then when the demand for a currency rises, the price of that currency, or *its exchange rate*, must rise
- And when the demand for a currency falls, its exchange rate falls



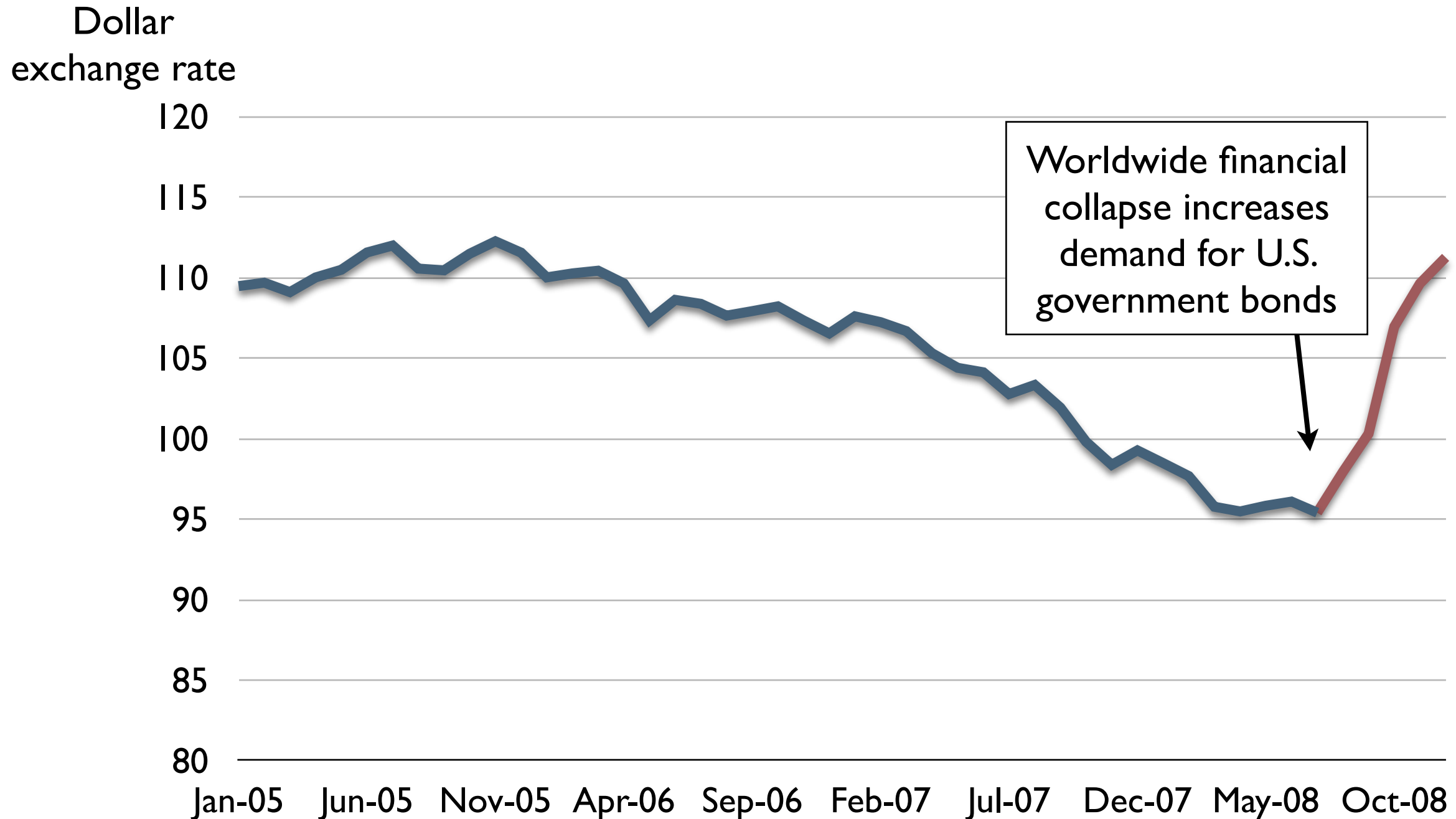
How does higher dollar demand affect the economy?

- If the demand for dollars increases, it must increase the price of dollars
- The dollar appreciates because the **nominal exchange rate** rises
- In the short run, price levels remain fixed, and the **real exchange rate** rises also
- When the real exchange rate rises, demand for U.S. exports will fall because the “real price” of U.S. goods has risen
- The real exchange rate is $RER = E P / P^w$. When that rises above 1, it's not profitable to export U.S. goods abroad because their exchange-rate-adjusted price, $E P$, is too high: $E P > P^w$
- U.S. demand for imports rises because they're a better deal
- When U.S. exports fall because they're more expensive, and U.S. imports rise because they're cheaper, U.S. net exports, NX , will fall, and Aggregate Demand falls

The U.S. real interest rate R affects U.S. net exports

- When the domestic real interest rate, R_t , **rises**:
 - Savers want to move their savings to the U.S., so they demand dollars
 - The dollar appreciates; the nominal and real exchange rates **rise**
 - Net exports **fall** because the real price of U.S. goods, the real exchange rate, has risen, so foreigners demand fewer U.S. exports and residents demand more U.S. imports
- When the domestic real interest rate, R_t , **falls**:
 - Savers want to move their savings abroad, so they demand fewer dollars
 - The dollar **depreciates**; the nominal and real exchange rates **fall**
 - Net exports **rise** because the real price of U.S. goods, the real exchange rate, has fallen, so foreigners demand more U.S. exports and residents demand fewer U.S. imports
- Changes in foreign real interest rates have similar (opposite) effects!

Exchange rates can also react to fear:
Historically, investors flocked to dollar assets *for safety*



What is new about our model economy?

Net exports respond to interest rates

- Because international savers respond to changes in real interest rates, net exports will be *negatively* related to the domestic real interest rate, R , and *positively* related to the world real interest rate, R^w
- When **R rises**, savers buy up U.S. stocks and bonds, driving up the value of the dollar, which raises the real price of U.S. exports, lowers the real price of U.S. imports, and thus **lowers NX**
- When **R^w rises**, the *reverse* happens because savers shift money abroad, which **raises NX**
- NX looks a little like the formula for domestic investment:

$$\frac{NX_t}{\bar{Y}_t} = \bar{a}_{nx} - \bar{b}_{nx}(\underline{R_t - \bar{r}}) + \bar{b}_{nx}(\underline{\bar{R}^w - \bar{r}}).$$

What does the IS Curve look like?

Recall that

$$Y_t = C_t + I_t + G_t + NX_t$$
$$\frac{Y_t}{\bar{Y}_t} = \frac{C_t}{\bar{Y}_t} + \frac{I_t}{\bar{Y}_t} + \frac{G_t}{\bar{Y}_t} + \frac{NX_t}{\bar{Y}_t}$$

What does the IS Curve look like?

Recall that $Y_t = C_t + I_t + G_t + NX_t$

$$\frac{Y_t}{\bar{Y}_t} = \frac{C_t}{\bar{Y}_t} + \frac{I_t}{\bar{Y}_t} + \frac{G_t}{\bar{Y}_t} + \frac{NX_t}{\bar{Y}_t}$$

$$\begin{aligned} 1 + \tilde{Y}_t &= \bar{a}_c + \bar{a}_i - \bar{b}_i (R_t - \bar{r}) + \bar{a}_g \\ &\quad + \bar{a}_{nx} - \bar{b}_{nx} (R_t - \bar{r}) + \bar{b}_{nx} (\bar{R}^w - \bar{r}) \end{aligned}$$

What does the IS Curve look like?

Recall that $Y_t = C_t + I_t + G_t + NX_t$

$$\frac{Y_t}{\bar{Y}_t} = \frac{C_t}{\bar{Y}_t} + \frac{I_t}{\bar{Y}_t} + \frac{G_t}{\bar{Y}_t} + \frac{NX_t}{\bar{Y}_t}$$

$$1 + \tilde{Y}_t = \bar{a}_c + \bar{a}_i - \bar{b}_i (R_t - \bar{r}) + \bar{a}_g \\ + \bar{a}_{nx} - \bar{b}_{nx} (R_t - \bar{r}) + \bar{b}_{nx} (\bar{R}^w - \bar{r})$$

The new IS Curve: $\boxed{\tilde{Y}_t = \bar{a} - \bar{b} (R_t - \bar{r})}$ is the same as the old, but

$$\bar{a} = \bar{a}_c + \bar{a}_i + \bar{a}_g + \bar{a}_{nx} - 1 + \underline{\bar{b}_{nx} (\bar{R}^w - \bar{r})}$$

$$\bar{b} = \bar{b}_i + \underline{\bar{b}_{nx}}$$

so some responses are different

$$\tilde{Y}_t = \bar{a} - \bar{b} (R_t - \bar{r})$$

$$\bar{a} = \bar{a}_c + \bar{a}_i + \bar{a}_g + \bar{a}_{nx} - 1 + \underline{\bar{b}_{nx} (\bar{R}^w - \bar{r})}$$

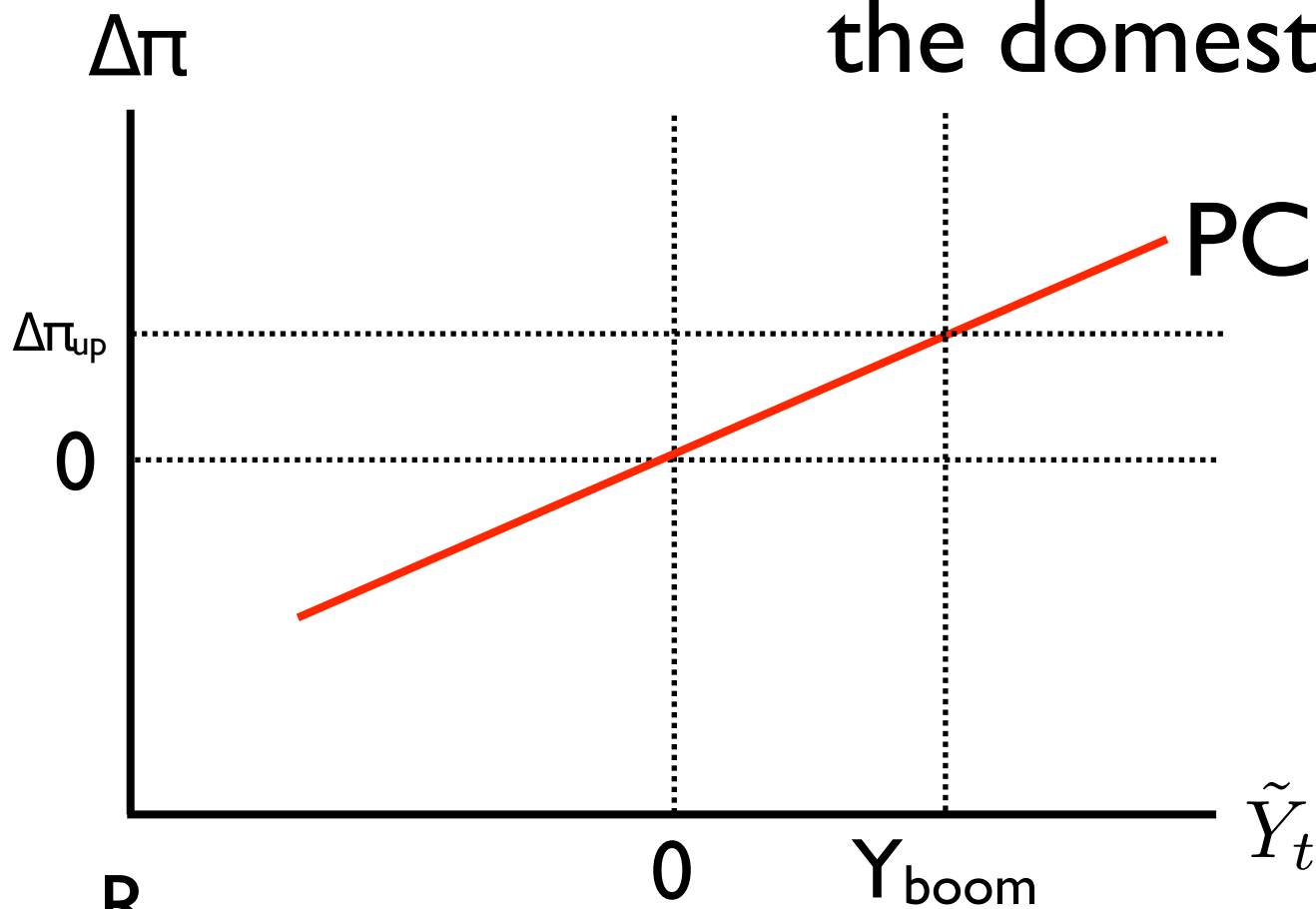
$$\bar{b} = \bar{b}_i + \underline{\bar{b}_{nx}}$$

- A rise in the world interest rate, R^w , will increase $\bar{a}$ and shift the IS Curve out, just like anything else that shifts $\bar{a}$
- Like other demand shocks, a rise in the world interest rate must be temporary
- Tricky: The IS curve is now **flatter**, because net exports also respond to the real interest rate, in addition to investment responding
 - Intuition: When R rises, investment I declines, *and so does* NX because the real exchange rate, the real price of U.S. goods, has risen
 - Math: The coefficient $\bar{b}$ is now bigger, so $\tilde{Y}$ is bigger at every R

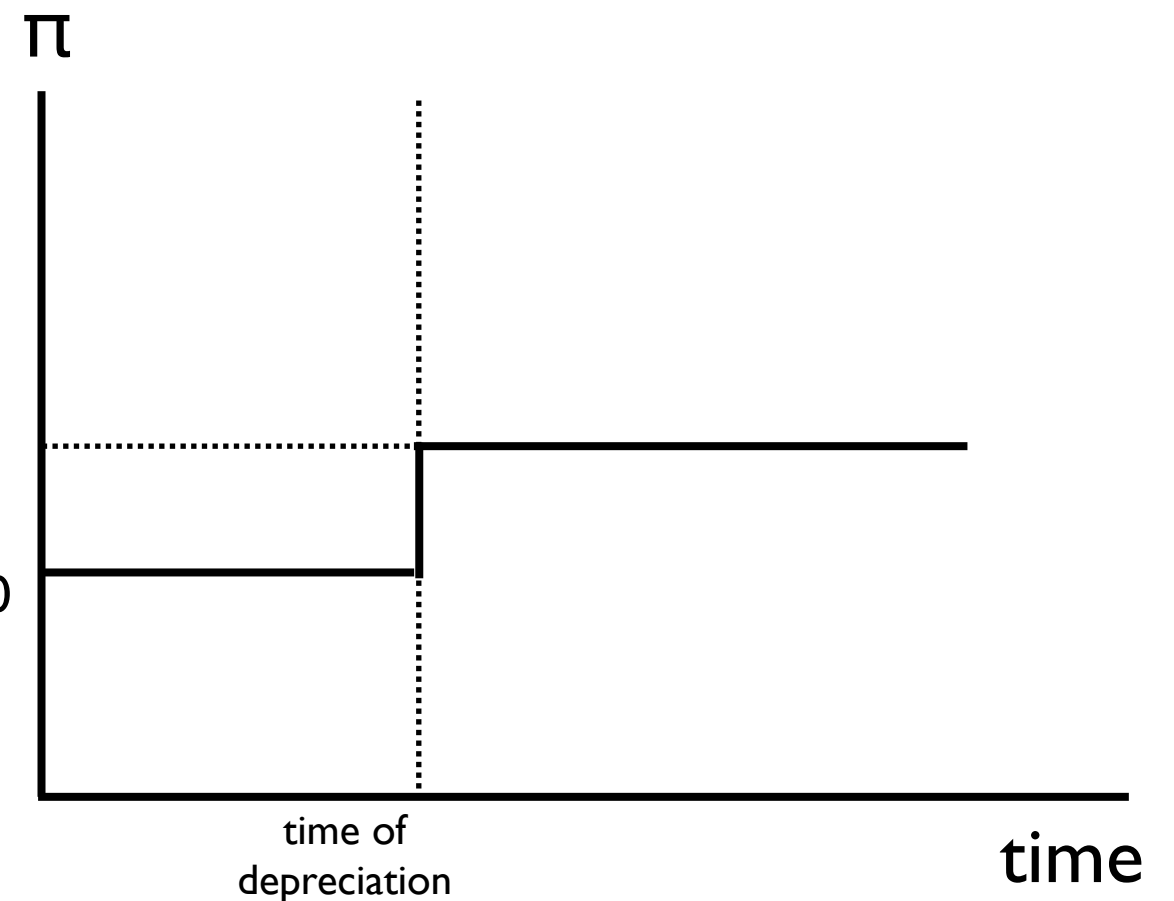
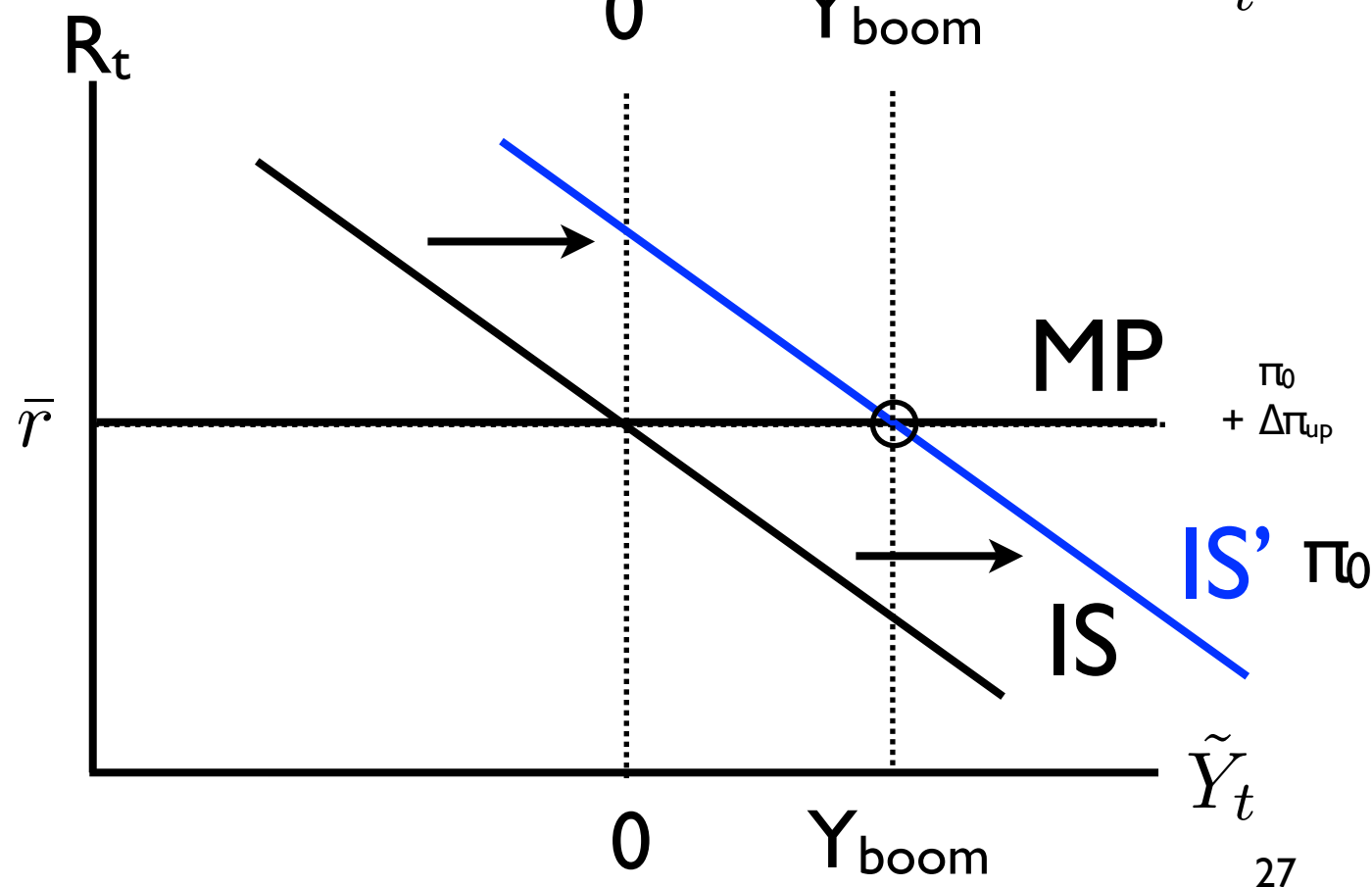
Let's examine a **rise** in the *foreign* real interest rate in the IS-MP-PC model

- Why would this happen?
- A foreign central bank might be fighting inflation by using its short-run model
- What do you think will happen to short-run output and inflation abroad?
- They should both *decline*
- What about short-run output and inflation here?
- Openness to investment and trade results in a mirror-image effect here: *higher output and higher inflation!*

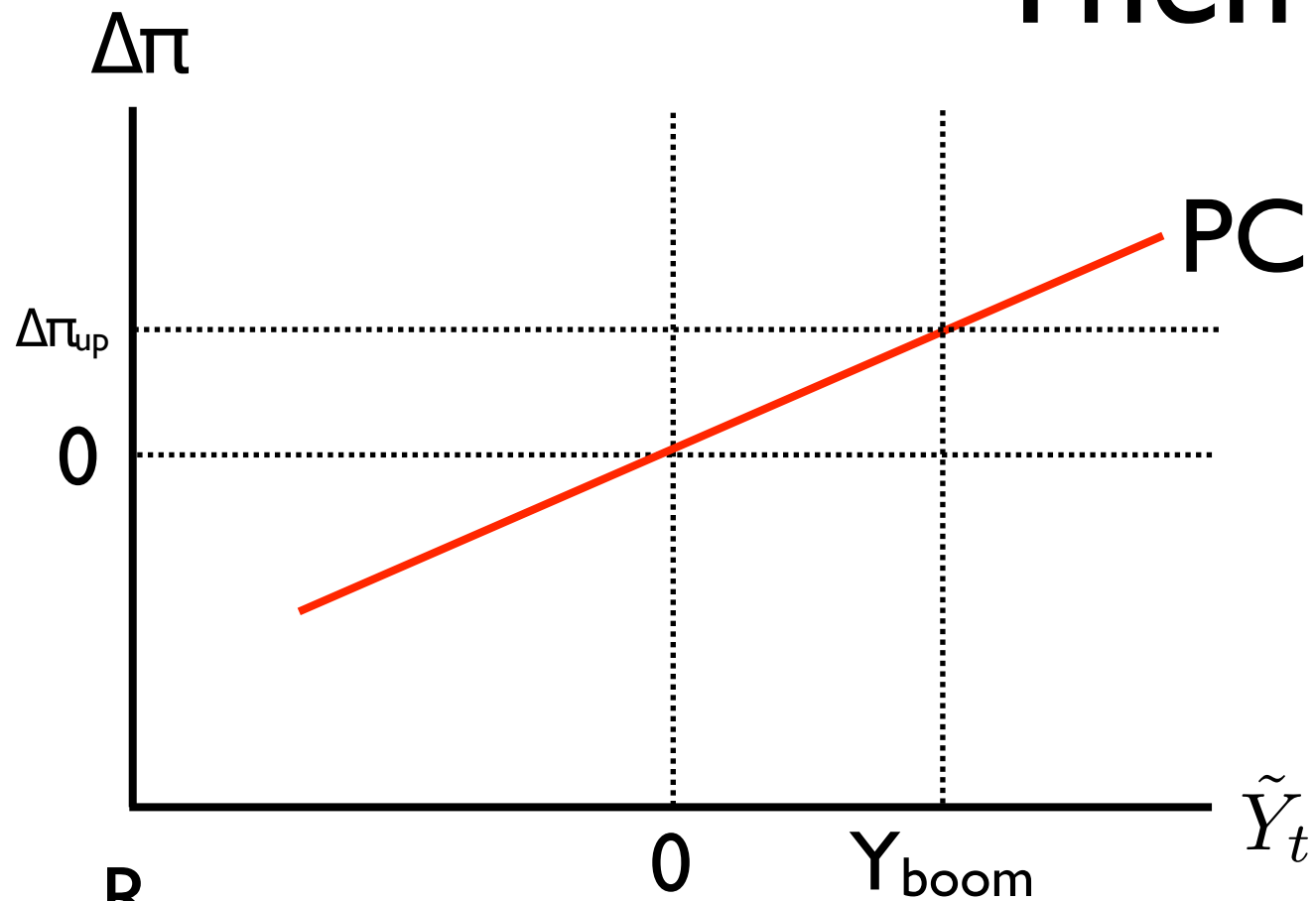
How does a rise in the *foreign* interest rate affect the domestic economy?



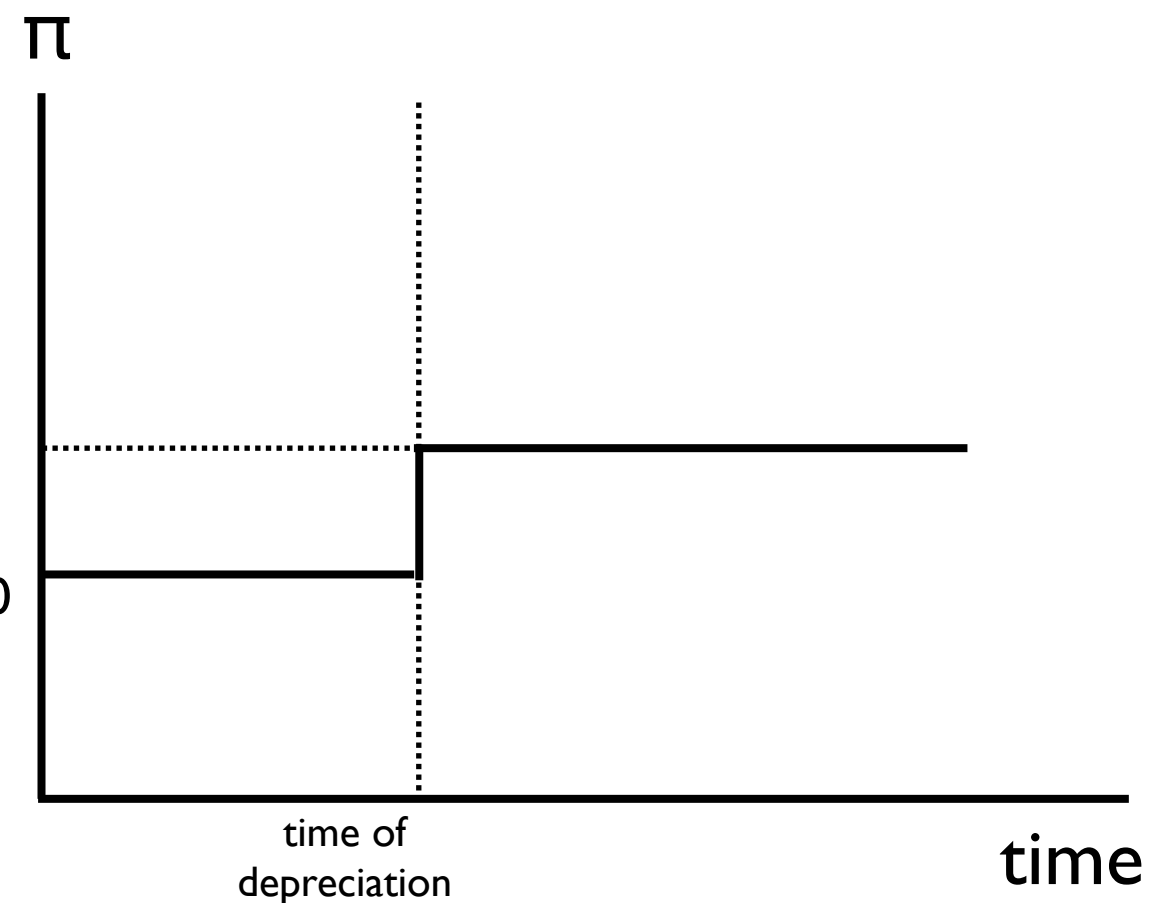
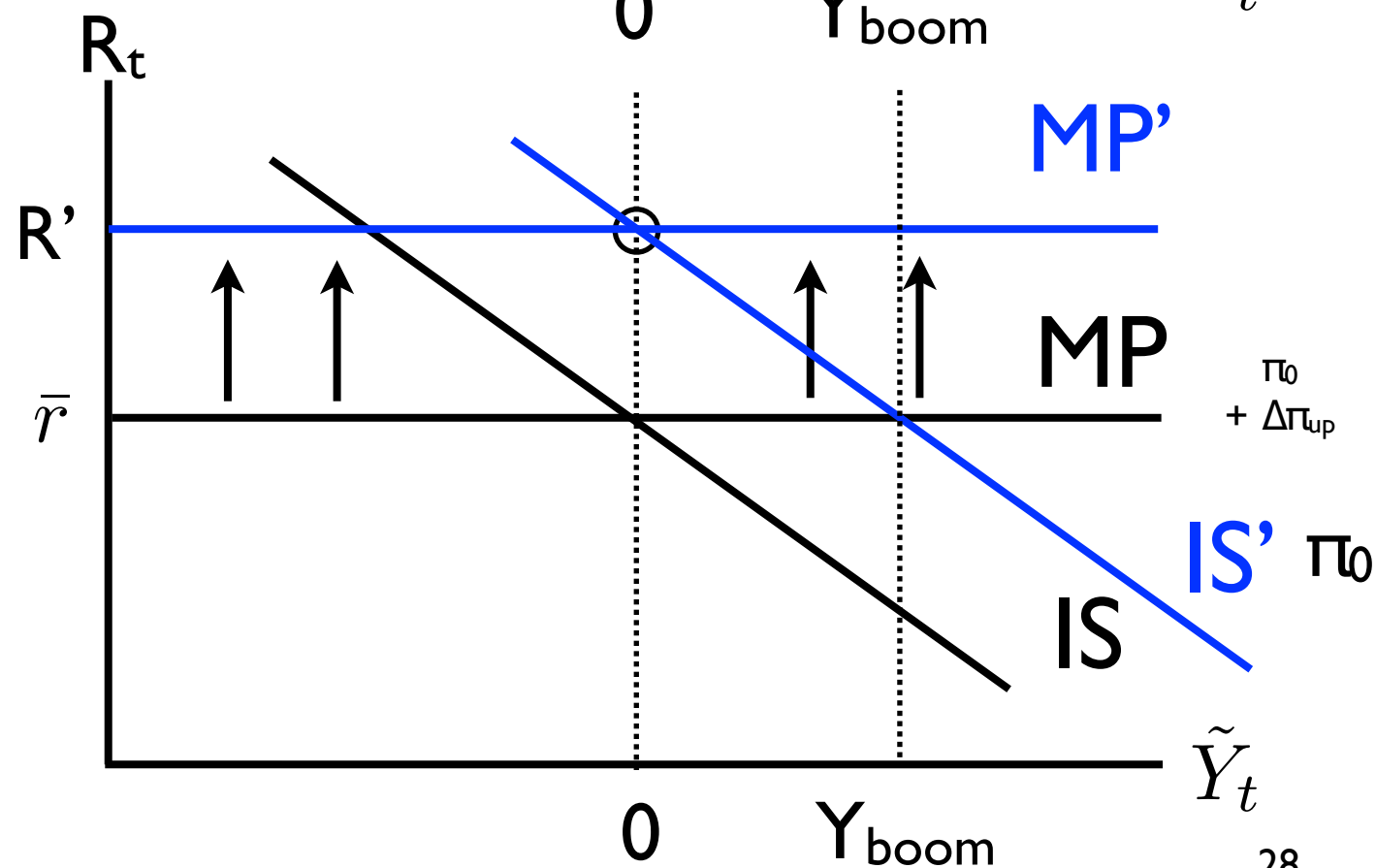
- If the foreign central bank raises its real interest rate, R^w , what do savers around the world do?
- Buy up more foreign assets, demanding fewer dollars, lowering the nominal exchange rate, E , and the dollar's real exchange rate. What does that do to U.S. net exports NX ?
- Increases them because they're cheaper! The IS curve shifts out, and the economy booms!
- Through the Phillips Curve, inflation rises!



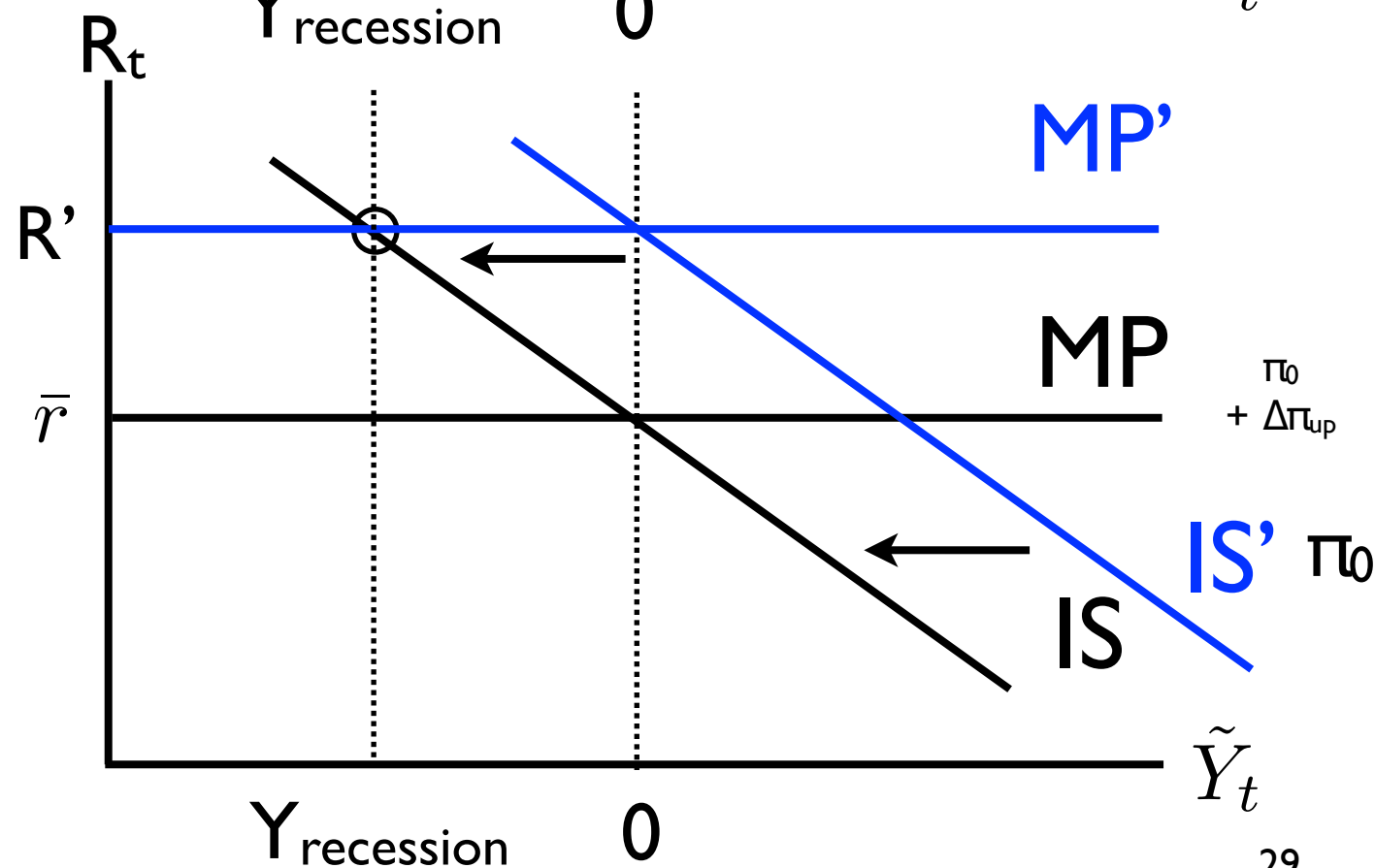
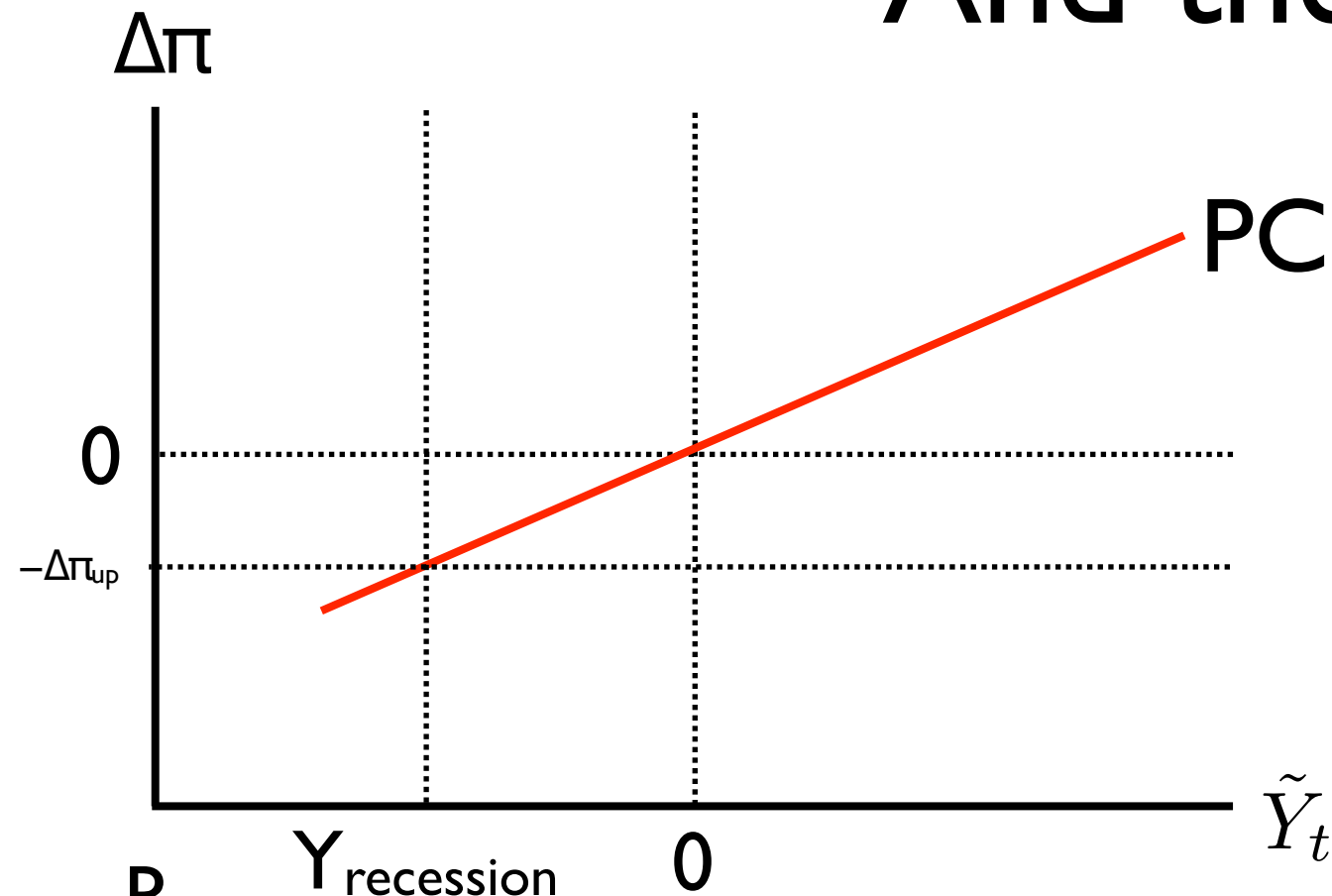
Then what?



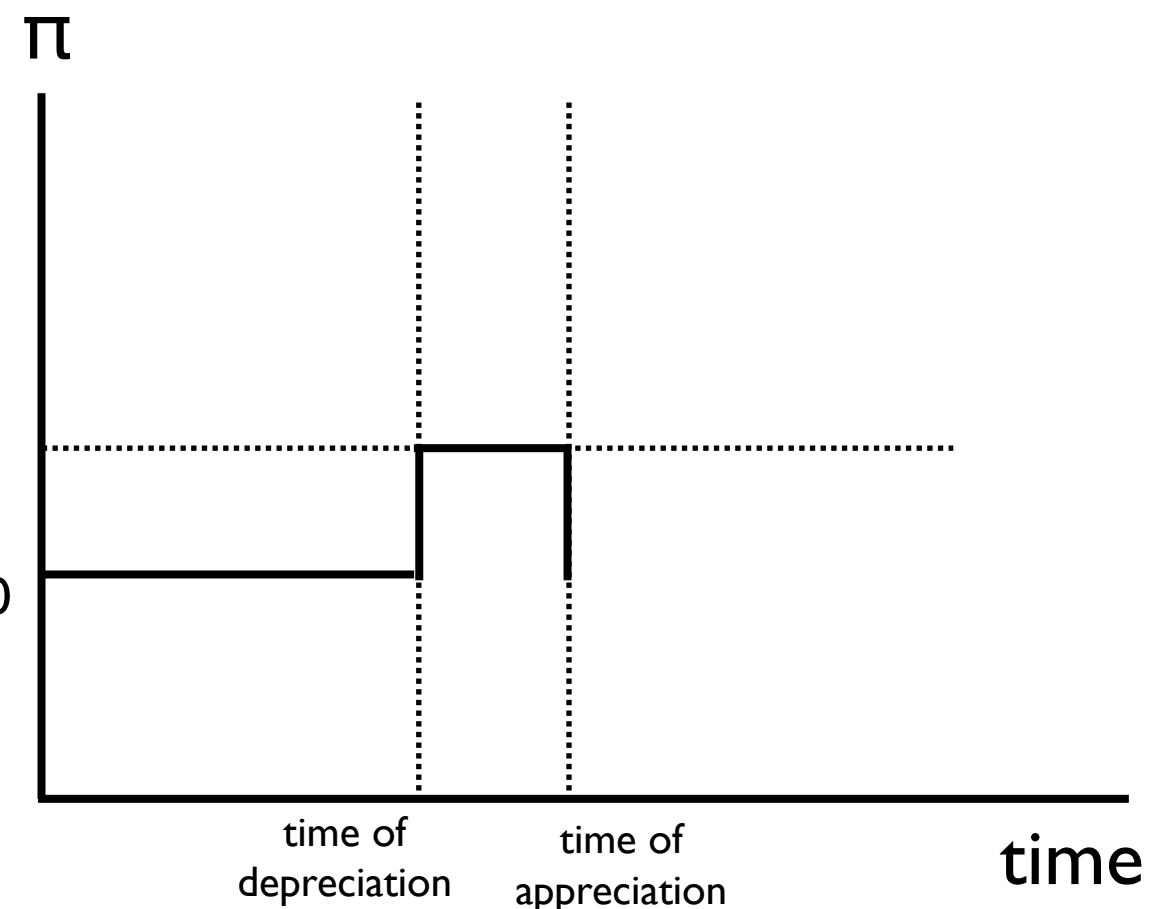
- In response to the rise in inflation above π_0 , the U.S. Federal Reserve will raise interest rates to combat inflation
- The MP curve shifts upward
- If the increase is large enough, this will stop the rise in inflation



And then what?



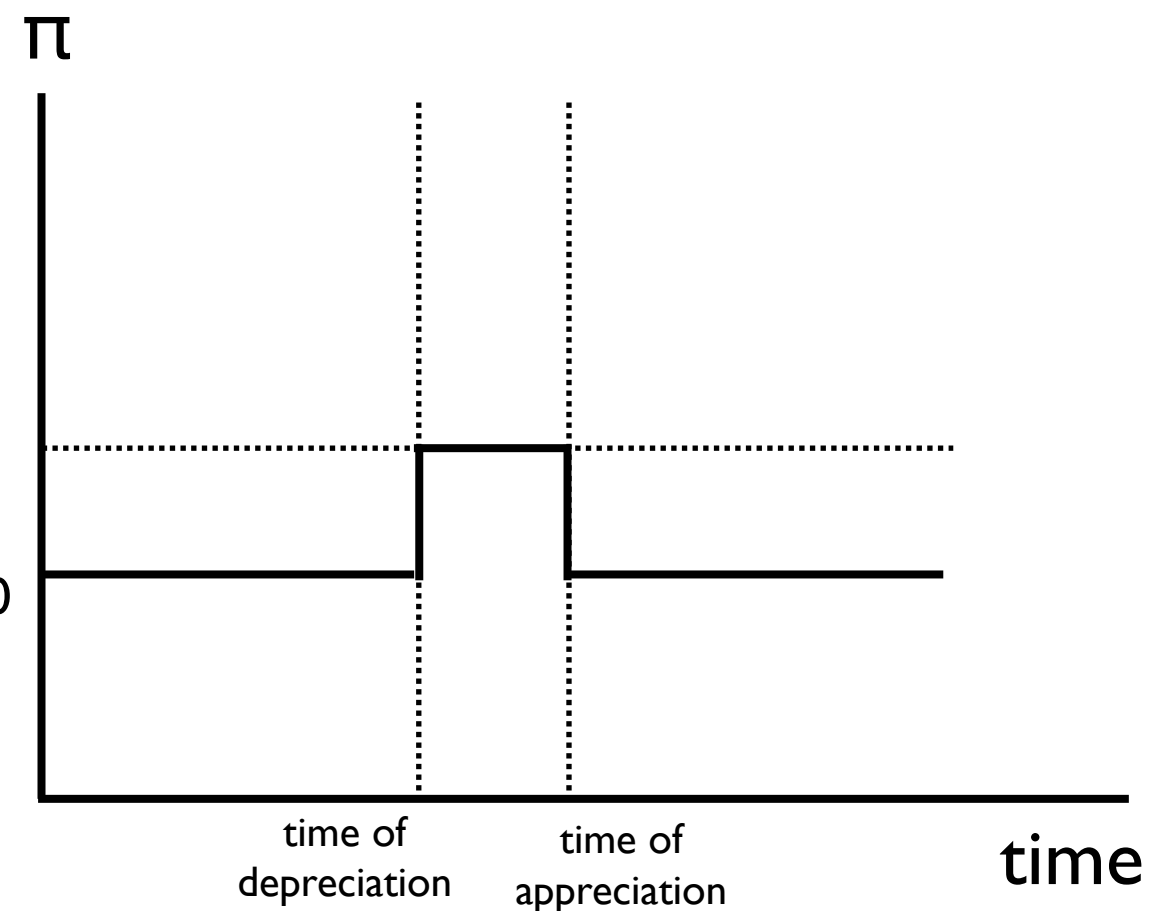
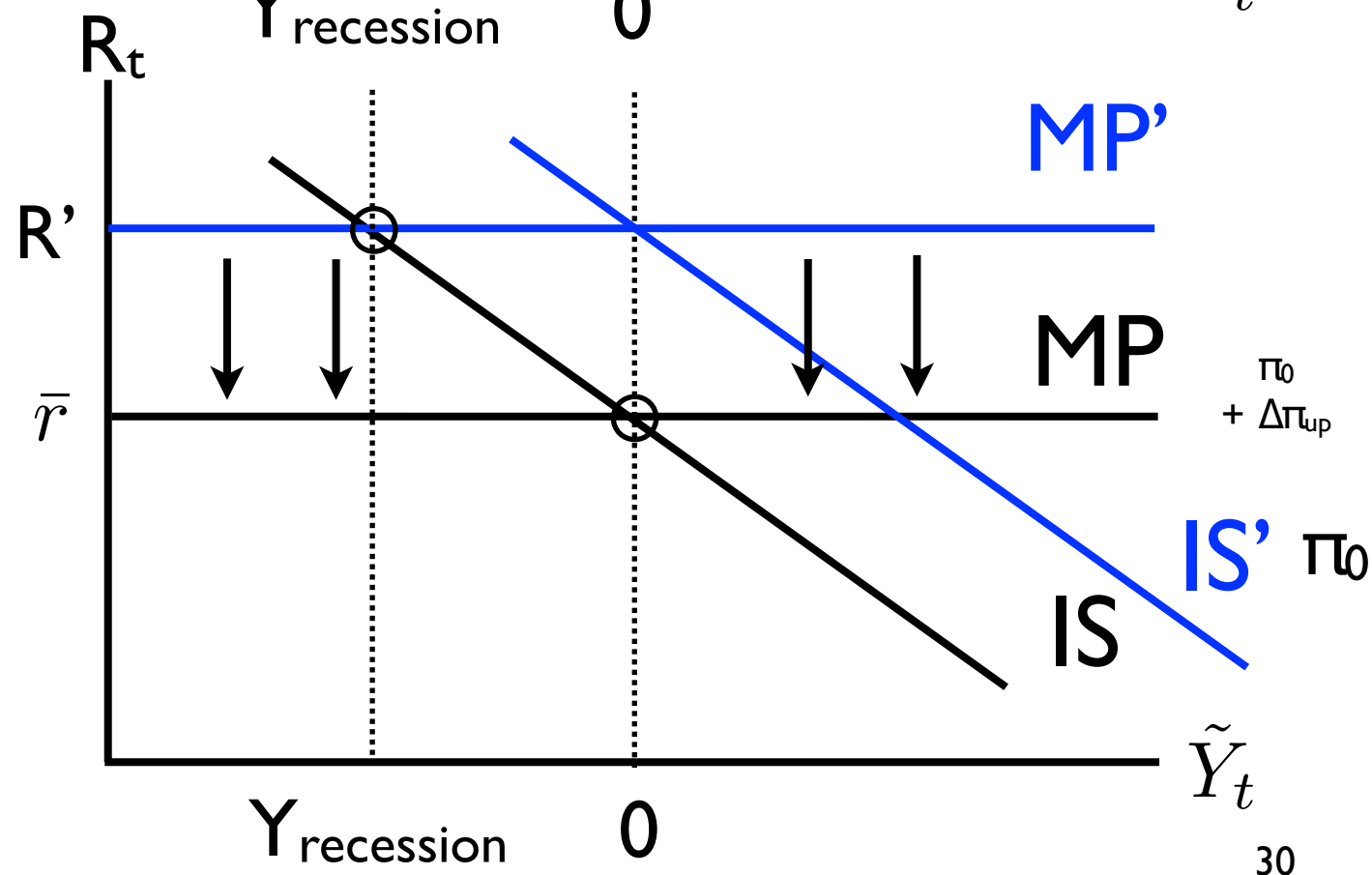
- Eventually, the positive shock to aggregate demand must wear off; Net Exports (NX) will return to normal —
 - Perhaps because the Fed has raised U.S. interest rates and caused an appreciation
- The IS curve shifts back leftward, causing a recession because the interest rate is high
- This *lowers inflation* through the Phillips Curve



Then the Fed can lower interest rates

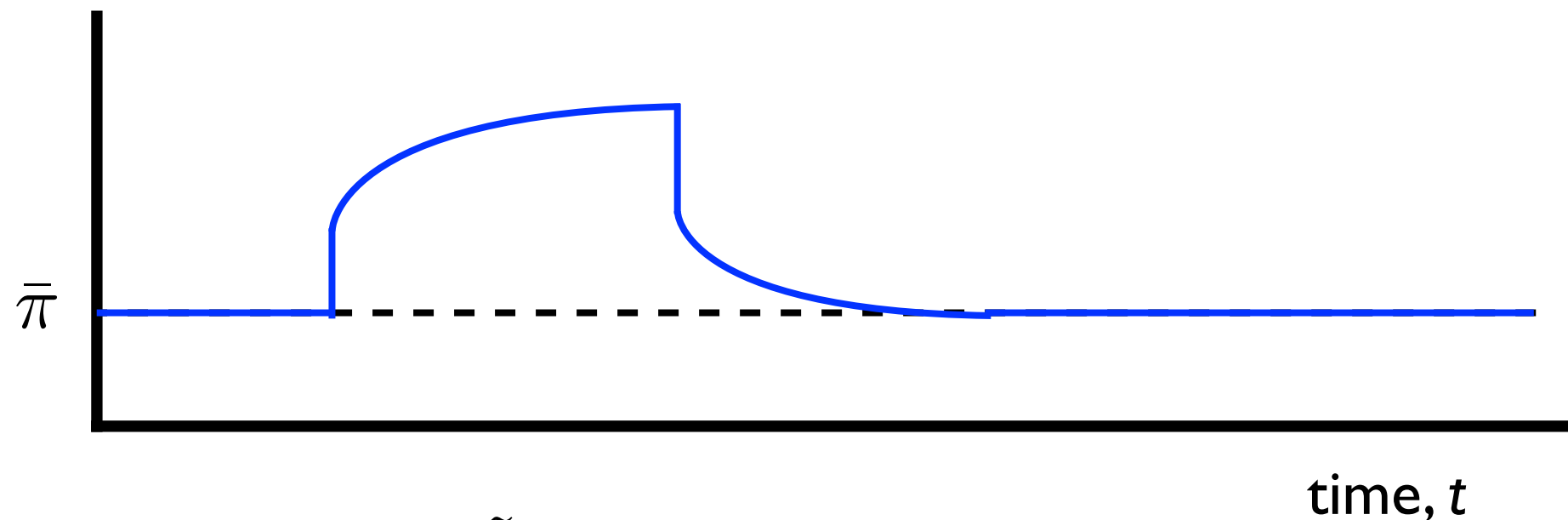


- Once the shock has worn off and inflation has returned to its normal level, the Fed lowers interest rates back to normal
- The MP curve falls, and we're back where we started, at short-run output equal to zero, neither a boom nor a bust
- Through the Phillips curve, inflation is stable

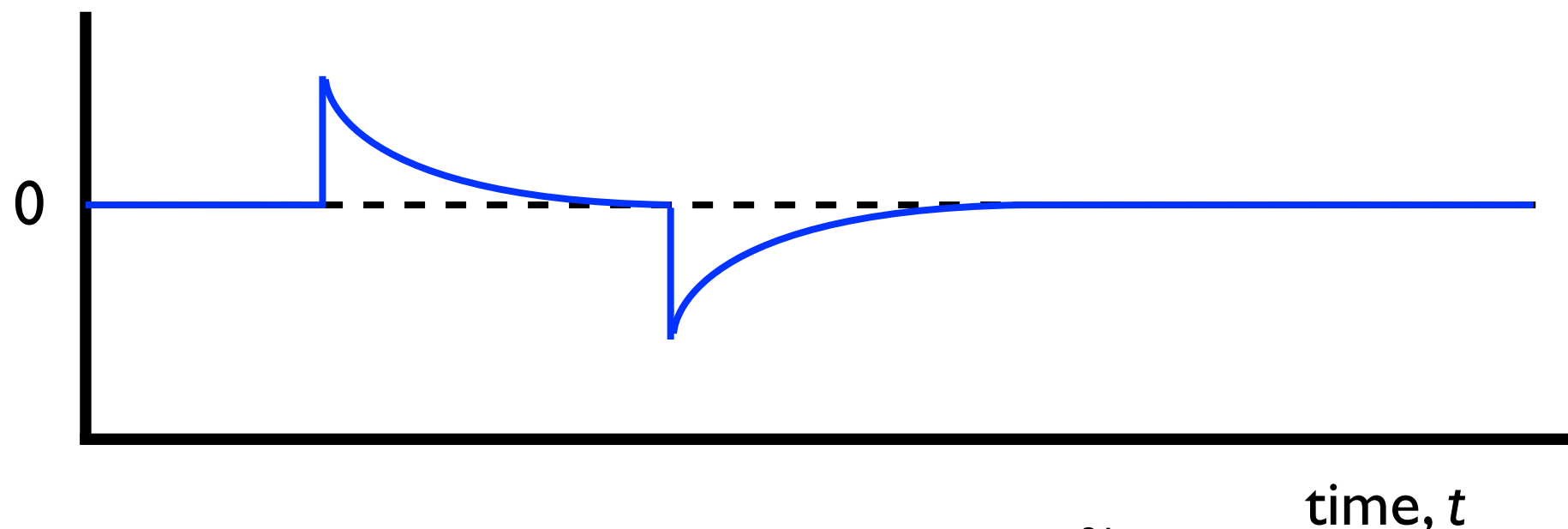


The time path of output and inflation after the temporary Aggregate Demand shock from foreign monetary policy

Inflation, π_t



Short-run output, $\tilde{Y}_t$



- Before the foreign central bank raises R^w , inflation is stable and output is growing at potential
- R^w rises, dollar demand drops as savings flow abroad, the real exchange rate falls, U.S. exports increase and imports decrease, and the IS curve shifts out, raising output and inflation *immediately*
- The Fed raises the MP curve, halting the increase in inflation and lowering short-run output back to 0
- When R^w returns to normal, the IS curve shifts back, *im* lowering output and inflation
- Inflation falls, and the Fed lowers interest rates, raising output